

IRIS FLOWER CLASSIFICATION USING MACHINE LEARNING

ADVANCED COMPUTER PROGRAMMING

GROUP 5

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PROJECT INTRODUCTION

- APPLY MACHINE LEARNING TECHNIQUES TO CLASSIFY IRIS FLOWERS.
- USE THE IRIS DATASET PROVIDED BY SCIKIT-LEARN.
- COMPARE DIFFERENT CLASSIFICATION ALGORITHMS.

DATASET

- Total samples: 150
- Number of features: 4
- Number of classes: 3

Species:

- Setosa

Versicolor

- Virginica

Dataset: Iris Flower Dataset

The Iris dataset contains 150 samples of iris flowers from three species: Setosa, Versicolor, and Virginica.

Iris Setosa



- Petal length: shorter
- Petal width: wider
- Easiest to classify

Iris Versicolor



- Petal length: medium
- Petal width: medium
- Between Setosa and Virginica

Iris Virginica



- Petal length: longer
- Petal width: narrower
- Most similar to Versicolor

DATASET

Features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Dataset: Iris Flower Dataset

The Iris dataset contains 150 samples of iris flowers from three species: Setosa, Versicolor, and Virginica.

Iris Setosa



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Iris Virginica



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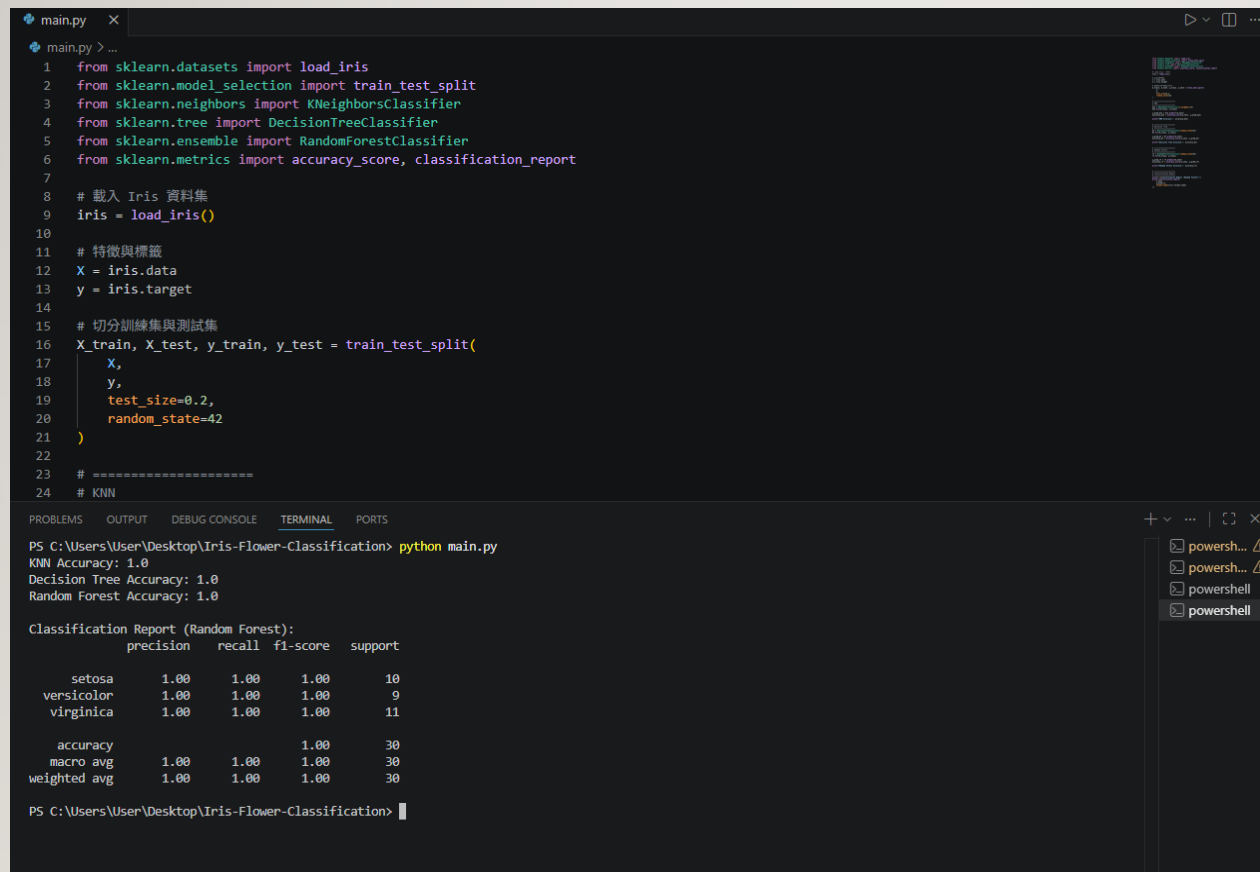
MACHINE LEARNING MODELS

- KNN
- Decision Tree
- Random Forest

DATA PREPROCESSING

- Training set: 80%
- Testing set: 20%
- `random_state = 42`
- `train_test_split()`

DATA PREPROCESSING



```
main.py x
main.py > ...
1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.neighbors import KNeighborsClassifier
4 from sklearn.tree import DecisionTreeClassifier
5 from sklearn.ensemble import RandomForestClassifier
6 from sklearn.metrics import accuracy_score, classification_report
7
8 # 載入 Iris 資料集
9 iris = load_iris()
10
11 # 特徵與標籤
12 X = iris.data
13 y = iris.target
14
15 # 切分訓練集與測試集
16 X_train, X_test, y_train, y_test = train_test_split(
17     X,
18     y,
19     test_size=0.2,
20     random_state=42
21 )
22
23 # =====
24 # KNN
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\User\Desktop\Iris-Flower-Classification> python main.py

KNN Accuracy: 1.0
Decision Tree Accuracy: 1.0
Random Forest Accuracy: 1.0

Classification Report (Random Forest):

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

PS C:\Users\User\Desktop\Iris-Flower-Classification>

IMPLEMENTATION

- Libraries: Scikit-Learn
- Functions:
 - `fit()`
 - `predict()`
 - `classification_report()`
 - `accuracy_score()`
- Insert main.py screenshot here

RESULTS

- Accuracy:
- KNN: 100%
- Decision Tree: 100%
- Random Forest: 100%

- Insert VS Code results screenshot here

CONCLUSION

- Successfully classified iris flowers.
- Compared three models.
- All models achieved 100% accuracy.
- Improved understanding of machine learning workflows.